

[Tree-like Causal Latency Model (TCLM) & Dual-Pulse Protocol] From Observational Latency Deconstruction to Interstellar and Intergalactic Deep-Space Chronometric Engineering

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Abstract

This paper aims to establish a complete theoretical framework for spacetime causal reconstruction and its accompanying deep-space chronometric engineering standards. It points out that the deduction in special relativity where "superluminal speeds ($v > c$) lead to time reversal" is essentially a misattribution of observational signal propagation latency as a reversal of physical time. This causes the Lorentz transformation to produce numerical overflow (imaginary numbers and negative time) under limit conditions.

This paper proposes the "Tree-like Causal Latency Model (TCLM)", strictly decoupling time into a "Topological Causal Order (Topological Trunk)" and a "Geometric Time-Flow Variance (Geometric Branches)". The topological trunk (t_{trunk}) is not a Newtonian universal absolute clock, but a strictly unidirectional topological causal order operator ($\text{sgn}(\mathcal{C}) \equiv +1$) that describes the underlying event settlement of the universe. The geometric branches represent the Proper Time and geometric observational latency of local physical entities under specific gravitational potentials and velocity fields.

To advance this theory into engineering practice, this paper simultaneously proposes a top-level communication standard: the "Dual-Pulse Protocol". This protocol breaks the traditional master-slave synchronization paradigm that uses specific celestial bodies (such as Earth) as privileged reference frames, establishing a **Decentralized Trunk Anchoring** mechanism. All nodes (whether Earth, Mars, or deep-space vehicles) act as equal branch observers. The protocol performs a dual-layer logical decoupling of the "local time flow rate (Γ)" and the "global initial constant (t_0)": On the local end, it establishes at least two sets of independent pulse units of different baseline lengths. Each set contains a quantum phase feature field post-projection imprint and a classical electromagnetic return pulse. Through multi-baseline cross-correlation operations and spacetime vector triangulation, it calculates the local branch curvature at a single point. On the deep-space end, only an ultra-low frequency (or single) initial handshake pulse is required to lock the initial integration constant t_0 . To cope with dynamic, continuously floating time fields, the protocol further introduces mechanisms for "Dynamic Pulse Frequency Scaling", "Continuous Curvature Quadratic Compensation", and "Causal Backpressure Buffering". Through the Worldline Path Integration of the spacetime worldline, the global objective causal order (t_{trunk}) is independently restored in dynamic gravity wells and non-inertial frames, achieving an engineering leap from "intergalactic relative calibration" to "global

causal order reconstruction". Combined with the astronomical data calibration of surface GPS (+38 $\mu\text{s/day}$) and the Earth-Mars system (47.9 $\mu\text{s/day}$), it is verified that under multi-frequency dispersion compensation, the theoretical resolution of this protocol is limited only by the hardware limits of current optical lattice clocks ($\mathcal{O}(10^{-16} \text{ s})$). The engineering alignment error budget converges to the nanosecond level ($\mathcal{O}(10^{-9} \text{ s})$), naturally transforming into a "Chronometric Gravitational Wave Detection Network" and forming a complete closed loop from theoretical causal deconstruction to deep-space spacetime engineering.

Epistemological Disclaimer

This theory adopts Information-theoretic Physics and topological geometry to structurally describe the causal chains of physical spacetime. The abstract concepts used in this paper, such as "static topological global map," "restricted read order," and "information propagation latency," are strictly confined to the realms of information theory and mathematical modeling. They are intended to provide an objective deconstruction of physical observation and do not make any metaphysical assumptions or extensions regarding the ontological entity of the universe.

Chapter 1: The Mathematical Boundary Failure and Domain Overflow of Special Relativity

Traditional special relativity faces two core mathematical collapses when dealing with extreme speeds, both rooted in "forcibly equating the electromagnetic propagation limit (c) with the ontology of cosmic time":

1. **The Negative Number Fallacy of Observed Time (t'):** In the transformation equation $t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$, when $v > c$, the numerator yields negative time. Traditional interpretations directly declare this as "time reversal". In essence, under geometric limit calculations, this is a misinterpretation of "Wavefront Reversal" as a physical reversal of time itself.
 2. **The Imaginary Number Fallacy of Proper Time (τ):** In the proper time formula $\tau = t\sqrt{1 - \frac{v^2}{c^2}}$, when $v > c$, it calculates the imaginary number i . This is an "Undefined Domain" overflow where the mathematical model exceeds its applicable boundaries, rather than a state where the physical structure of time dissolves into nothingness.
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Qualitative Assessment of Boundary Failure

When the Lorentz transformation yields imaginary and negative times at $v > c$, it has traditionally been interpreted directly as "physically impossible, therefore superluminal speeds do not exist". However, this interpretation conflates two distinct concepts:

- **Mathematical model failure:** The input to the formula exceeds its valid range, thus outputting an invalid value.
- **Physical entity impossibility:** Natural laws prohibit the existence of the phenomenon based on first principles.

This paper points out that the imaginary and negative times under $v > c$ only prove that the Lorentz transformation fails in that region; they do not constitute a physical negation of superluminal speeds. This conflation is the logical root of the time travel paradox that has persisted for a century.

Chapter 2: Core Postulates and Global Causal Equations of the Tree-like Causal Latency Model (TCLM)

1. Core Postulates

To repair the mathematical collapse of relativity under limit conditions, we establish the following postulates to strictly decouple the "topological causal order (trunk)" from the "relative observational latency (branches)" in the physical world:

- **Postulate 1: Topological Causal Order & Branch Curvature**
 - **Topological Causal Trunk (Topological Causal Order):** Represents the underlying, objective, irreversible causal sequence of the universe (t_{trunk}). **It is specifically emphasized that t_{trunk} is not a Newtonian Universal Clock ticking uniformly throughout universal space, nor does it provide any Preferred Simultaneity Slice.** t_{trunk} is a topological causal scalar parameter solely used to index the sequential settlement order of events (Directed Acyclic Graph, DAG). The existence and event settlement of all physical entities in the universe are equally rooted in this causal order, progressing strictly unidirectionally along the trunk.
 - **Geometric Relative Branches & Curvature:** Represent the relative observational trajectories and local Proper Time (τ) of various physical entities growing from the causal trunk. The essence of relative time flow rates is the curvature of the branches on the four-dimensional geometric manifold. Relativistic effects measured by local observers (such as time dilation and gravitational redshift) are all results of path integrals along geometric branches.
 - **Dynamic Environmental Adaptability (Continuous Curvature Field Fluctuation):** Physical entities are not locked into the hierarchical structures of specific celestial bodies. Spacetime curvature is inherently a continuously floating field that changes in real-time according to the current gravitational potential and velocity field. When an object undergoes intergalactic movement, its intrinsic branch directly adapts to the local continuous geometric curvature. Only "relative time differences induced by the current geometric curvature" exist between objects, and they collectively share the same global causal order.
 - **Forward Growth Constraint:** Under the standard topological causal operator ($\text{sgn}(\mathcal{C}) \equiv +1$), the growth vectors of all branches flow strictly in a single, forward direction. The topological causal operator is strictly positive-definite and absolutely forbids backward growth upstream into the trunk within a single manifold. Historical coordinates belong to a topologically irreversible state, completely eliminating time reversal and paradox backdoors.
- **Postulate 2: Decoupling of Event Ontology and Signal Carrier (Isomorphic Model of Thunder and Lightning)**
 - **Decoupling of the Event Ontology and Lightning:** Using "the existence of the thunder/lightning event ontology" and "the outward propagating lightning light" as an isomorphic model: The instant a thunder event is settled on the topological trunk (t_{trunk}) is the settlement point of that physical entity in the topological causal order; whereas the lightning light received by the observer is merely the geometric branch signal of the event propagating outward ($t_{\text{observed}} = t_{\text{trunk}} + \Delta t_{\text{latency}}$).
 - **Limit Propagation and Division of Time Ontology:** Lightning is only a carrier for information transmission (geometric branch); its propagation latency does not equal the occurrence time of

the thunder event itself. Even if one masters a propagation method exceeding the speed of light (perceiving the thunder event faster than the light), it only means the signal transmission latency is shortened (the branch is flattened). It absolutely does not mean the observer travels backward to before the thunder event occurred (causal settlement).

2. Global Causal Equation and Dual-Layer Geometric Decoupling

The TCLM global causal equation adopts a dual-layer modular architecture, decoupling macroscopic topological causal relations from microscopic geometric field latency mechanisms:

$$t_{\text{observed}} = t_{\text{trunk}} + \Delta t_{\text{latency}}$$

Where the microscopic geometric field operator (latency mechanism) is defined as:

$$\Delta t_{\text{latency}} \equiv \int_{\mathcal{P}} \frac{\sqrt{g_{ij}^{\text{branch}} dx^i dx^j}}{v_{\text{signal}}(x)}$$

(The t_{trunk} parameter satisfies strictly unidirectional increments; the integral term is determined comprehensively by geometric distance, propagation speed, and branch curvature, always being positive, with units in seconds).

Parameter Symbol Explanations:

- t_{observed} (Observed Time): The scalar time when the observer receives the information afterimage in a specific coordinate system.
- t_{trunk} (Topological Causal Order Parameter): The underlying causal settlement scalar of event occurrence, satisfying the strict unidirectional increment postulate ($\frac{dt_{\text{trunk}}}{d\tau} > 0$). It represents the causal topological position of the event, not a globally synchronized physical clock.
- $\Delta t_{\text{latency}}$ (Geometric Latency Amount): The geometric latency accumulated by the information propagating in the branch space manifold, measured in seconds (s), and is always a positive definite value ($\Delta t_{\text{latency}} \geq 0$).
- $\mathcal{P}$ (Propagation Trajectory): The physical geometric path traversed by the information carrier in the three-dimensional branch space manifold.
- g_{ij}^{branch} (Branch Space Metric Tensor): A positive definite geometric tensor describing local spatial curvature, gravitational potential, and the ultimate resistance to information propagation ($i, j \in \{1, 2, 3\}$).
- dx^i, dx^j (Differential Spatial Displacement): The minute coordinate variations of the information carrier advancing along the spatial propagation path.
- $v_{\text{signal}}(x)$ (Local Effective Propagation Speed): The actual propagation rate of the information carrier at coordinate point x (e.g., the effective speed of light for electromagnetic waves in the local field, satisfying $v_{\text{signal}}(x) > 0$), ensuring the equation has a rigorous time dimensionality $[\mathbf{m}]/[\mathbf{m}/\mathbf{s}] = [\mathbf{s}]$.

Chapter 3: Deep Space Engineering Implementation: Dual-Pulse Protocol and Trunk Causal Order Restoration

Preliminary Symbol Definitions

- Γ_{local} : The local time dilation factor of any node relative to the global causal trunk t_{trunk} , where $\Gamma \geq 1$.

- $\gamma_{B/A}$: The relative time rate of change between two observation endpoints A and B , $\gamma_{B/A} = d\tau_B/dt_A$. The two must not be conflated.

0. Engineering Necessity and Paradigm Shift of the Dual-Pulse Protocol

0.1 Engineering Motivation & Necessity

Traditional deep space synchronization architectures assume time flow rates to be predictable constants or static offsets. However, under the TCLM framework, the local time dilation factor (Γ) relative to the trunk is fundamentally a continuous floating field dynamically influenced by gravitational and velocity fields. This characteristic leads to three major bottlenecks in deep-space chronometric engineering, which constitute the engineering necessity for this protocol:

1. **Local Imperceptibility**: When a single observation node is located on a fixed celestial body (like Earth's surface), because the local curvature gradient changes are extremely minute, fluctuations in the time flow rate are almost imperceptible in short-term measurements. Observers easily mistakenly believe the time flow rate is a constant, thereby ignoring the dynamic corrections required for global alignment.
2. **Interplanetary Scale Amplification**: Once crossing over to a deep space endpoint (like the Earth-Mars system), minute differences in local curvature and potential energy will continuously accumulate and amplify during the time integration process. The time flow rate variance between Earth and Mars can reach the magnitude of $47.9 \mu\text{s}/\text{day}$, and this difference changes dynamically with orbital eccentricity, solar gravitational potential perturbations, and celestial perturbations; it cannot be extrapolated long-term using a single fixed constant.
3. **Infeasibility of Continuous Sync**: Attempting to rely on high-frequency, bidirectional signals for continuous phase-locked synchronization faces two limits: First, the **light-speed propagation latency limit** (e.g., one-way latency between Earth and Mars takes 3 to 22 minutes, making real-time feedback impossible); Second, the **unpredictability of continuous dynamic perturbations**, causing any fixed mathematical extrapolation to inevitably diverge over time.

0.1.1 Applicability Boundary

The Dual-Pulse Protocol is not intended to replace all deep-space synchronization architectures, nor is it necessary in all scenarios. Its application conditions are as follows:

1. **Sufficient Local Computing Power**: Nodes must be capable of executing cross-correlation calculations, geometric reconstructions, and continuous integrations. If local computing power is insufficient, traditional telemetry synchronization is adequate.
2. **Communication Interruption Tolerance**: If a mission allows for continuous, high-frequency handshakes and the communication link is stable, traditional DSN timing methods meet the needs. The primary value of this protocol lies in providing offline, autonomous chronometric capabilities.
3. **Long-Term Interplanetary or Beyond Missions**: For near-Earth or short-term missions where relative time flow rate differences are minimal, the engineering yield of introducing this protocol is limited.

In short, the Dual-Pulse Protocol is an offline time restoration solution designed for "long-term, offline, high-computing" deep-space nodes, rather than a universally required generic solution.

0.2 Paradigm Shift in Synchronization: From "Earth Model Dependency" to "Local Physical Measurement"

The fundamental limitation of traditional deep space synchronization (like the Deep Space Network) lies not only in its master-slave ideology treating Earth as a privileged reference frame, but also in its methodological complete reliance on astronomical formulas and ephemeris models from the Earth side. The deep space node itself does not have the capability to measure time flow rates and can only passively receive calibration parameters calculated by Earth. Once communication is interrupted or the ephemeris model fails, the node loses its temporal autonomy.

Under the TCLM framework, both Earth time (t_{Earth}) and Mars time (t_{Mars}) are merely "geometric branch afterimages" under local gravity wells and states of motion; neither serves as the objective time standard of the universe.

The fundamental breakthrough of the Dual-Pulse Protocol is empowering any deep-space node with the capability of "local physical measurement". Nodes no longer need to query Earth's gravitational field formulas, but instead directly measure their own geometric curvature relative to the global causal trunk (Γ_{local}) by comparing local quantum phase feature fields and classical pulses. The models calculated by Earth merely serve as initial references, not the sole source of truth.

Conceptual Dimension	Traditional Deep Space Sync (DSN-type)	Dual-Pulse Protocol (TCLM Topological-type)
Reference Frame Positioning	Earth privileged reference frame (Master-Slave)	Decentralized Nodes with no privilege
Alignment Target	Follows Earth time t_{Earth}	Independently restores global causal trunk t_{trunk}
Communication Dependency	Requires high-frequency/continuous bidirectional handshakes to maintain phase lock	Only requires initial value lock, followed by local differential autonomous calculation
Physical Essence	Engineering compromise to compensate for signal latency	Observer's decoding of the universe's underlying topological causal order

Based on the above paradigm shift, this protocol abandons traditional high-frequency bidirectional phase-locked architectures, performing a **dual-layer logical decoupling** of the "local time dilation factor (Γ)" and the "global initial constant (t_0)", establishing a multi-layered dynamic control system.

This paper uses the Earth-Mars system as the first-stage engineering calibration case. The applicability of the Dual-Pulse Protocol is not limited to interplanetary distances; provided that an initial handshake can be completed, its autonomous restoration and communication-interruption tolerance logically extend to interstellar and even intergalactic scales.

1. Layer One: Local Spacetime Vector Triangulation (Earth-to-Earth / Mars-to-Mars)

Engineering Definition of "Dual-Pulse"

The "dual-pulse" referred to in this protocol means that the local end establishes at least two reference baselines of different lengths and simultaneously triggers two sets of independent pulse units.

Each set of pulse units contains one quantum phase post-projection and one classical electromagnetic return. A single baseline can only provide a latency difference projection in a single dimension, which cannot uniquely solve for the geometric deviation angle of the spacetime manifold.

Therefore, the minimum requirement for the dual-pulse is two reference baselines of different lengths; if a third baseline of a different length is available, the uniqueness of the θ solution and measurement stability can be further verified through a closed traverse method.

Internal Operations of a Single Pulse Unit (Quantum-Classical Coupled Measurement)

For any independent baseline (endpoints L_1, L_2), its single pulse measurement consists of two indivisible physical steps, strictly adhering to the no-communication theorem:

- **Step One: Zero-Latency Quantum Correlation Imprint:**

Utilize a locally pre-entangled quantum system to synchronously trigger measurements at both ends L_1 and L_2 .

The single-sided reduced density matrix ($\rho = \frac{1}{2}I$) only presents random statistical eigenvalues and cannot transmit real-time information; however, this measurement leaves a "physical correlation imprint with zero transmission latency" at both ends across macroscopic distances.

- **Step Two: Classical Light Cone Latency Return:**

Along the same baseline, the remote end (L_2) encodes the post-read measured feature data stream (including phase imprints and occurrence sequences) into classical electromagnetic signals and transmits them back to the local end (L_1).

Its propagation trajectory constitutes a standard local light cone, providing a geometric latency reference across space.

Dual-Pulse Triangulation & Cross-Correlation Algorithm

The local end system must simultaneously collect pulse measurement data from two (or more) reference baselines.

The local end (L_1) performs a **Cross-Correlation** operation between its retained quantum measurement imprint sequence and the classical feature data streams transmitted from each baseline:

$$R_{AB}(\tau) = \int A_{L_1}(t) B_{L_2}(t + \tau) dt$$

Through continuous feature curve matching and local differential restoration by comparing two or more baselines of different lengths, the system can eliminate single-axis blind spots and precisely calculate the geometric angle θ between the local clock's worldline tangent vector u^μ and the global causal trunk normal vector n^μ .

The mapping relationship between the "local time dilation factor (relative to the trunk)" Γ_{local} and the geometric deviation angle θ is defined as:

$$\cos \theta = \frac{d\tau}{dt_{\text{trunk}}} = \frac{1}{\Gamma_{\text{local}}(t)} \implies \Gamma_{\text{local}}(t) = \sec \theta \geq 1$$

When local spacetime has no relative velocity or gravitational potential tilt, $\theta = 0^\circ$ and $\Gamma_{\text{local}} = 1$ (the local time flow rate equals the trunk limit); when a curvature tilt exists, the dilation factor $\Gamma_{\text{local}} \approx 1 + \frac{1}{2}\theta^2$.

This step is **purely conducted locally via multi-baseline data comparison and geometric reconstruction**, without relying on real-time intergalactic communication.

When the system possesses three or more sets of different length baselines, the uniqueness and stability of the θ solution can be further verified using a closed traverse method.

In strong gravitational fields or extreme curvature environments, the system must mandatorily invoke the exact solution $\Gamma_{\text{local}} = \sec \theta$ and must not use the small-angle expansion approximation.

1.1 Geometric Insufficiency of a Single Baseline

In the Dual-Pulse Protocol, the requirement of at least two baselines of different lengths is not an engineering preference; it is a necessary condition for angular measurement.

A single baseline can only provide a scalar latency difference:

$$R_{AB}(\tau) = \int A_{L_1}(t) B_{L_2}(t + \tau) dt$$

This scalar only tells the system that "some latency exists," but it cannot distinguish the source of that latency:

- Time-flow-rate difference
- Spatial-curvature difference
- Instrument-response difference
- Random fluctuation

More critically, if only a single length of accumulated latency is available, an unknown angle θ cannot be mathematically distinguished from multiple possible solutions.

Any system attempting to resolve a local geometric deviation angle with only a single baseline obtains merely a scalar projection along one axis. The solution space exhibits rotational degeneracy. Therefore, two or more independent baselines of different lengths are not an engineering preference of this protocol, but a physical necessity for uniquely resolving the geometric deviation angle through accumulated latency difference. Directional differences serve only as auxiliary measures when dealing with anisotropic perturbations or reducing multipath interference; the core necessary condition remains length difference.

2. Layer Two: Interplanetary Initial Handshake Pulse (Deep Space Initial Constant Lock)

Because the mathematical restoration equation for the global causal order inevitably contains an unknown **Integration Constant** (t_0), purely relying on local calculations of the rate of change $\Gamma(t)$ only reveals the amount of time advancement and cannot determine the initial alignment constant of both parties' time axes.

• **Single Handshake Execution Steps:**

1. The reference end (e.g., Earth) emits a single classical deep-space radio pulse to the target end (e.g., Mars), carrying the initial timestamp $t_{\text{Earth_start}}$.
2. The Mars end receives the pulse and calculates the light-speed latency $\Delta t_{\text{propagation}}$ by combining orbital geometric distances and multi-frequency dispersion compensation, establishing the initial alignment relationship (t_0) between Mars local time and Earth reference time.

- **Autonomous Operation State:** Once the initial value t_0 is locked, the Mars end can cease high-frequency deep-space communications and rely purely on the real-time $\Gamma_{\text{Mars}}(\tau)$ obtained from "Layer One Local Triangulation" to independently conduct autonomous restoration of the global causal order:

$$t_{\text{trunk_reconstructed}}(t) = t_0 + \int_{\tau_0}^{\tau} \frac{1}{\Gamma_{\text{Mars}}(\tau')} d\tau'$$

- **Blackout Re-synchronization Mechanism:** If a deep-space node enters a long-term communication blackout due to solar occlusion, the local continuous field integration engine can still independently track time flow rate changes without triggering a causal fault. However, the integration constant t_0 may accumulate drift, so upon exiting the blackout, a re-handshake pulse must be executed to relock the initial alignment constant. The geometric curvature measurement of the time flow rate itself is not invalidated by communication interruptions.

3. Layer Three: Continuous Flow Rate Dynamic Control (Adaptive Frequency Scaling, Quadratic Compensation, and Causal Backpressure)

Addressing issues of frequency asymmetry and data queue overflow caused by continuous dynamic floating of time flow rates ($\frac{d\Gamma}{dt} \neq 0$), the protocol introduces the following dynamic control mechanisms:

- **Dynamic Pulse Frequency Scaling:** Both ends calculate the current relative flow rate ratio $\lambda(t) = \frac{d\tau_{\text{target}}}{d\tau_{\text{source}}}$ through the initial handshake. The high flow rate end (source) automatically and dynamically adjusts its synchronization pulse emission period:

$$\Delta t_{\text{source}} = \frac{\Delta \tau_{\text{target}}}{\lambda(t)}$$

When the target end is in a low flow rate zone ($\lambda \ll 1$), the source end automatically extends the emission interval to match the target end's local processing frame rate, preventing the low flow rate end from being bombarded by high-frequency pulses.

- **Continuous Curvature Quadratic Compensation:** In continuously changing gravitational and acceleration fields, the dual-pulse signal header forcibly attaches the node's current "trunk dilation acceleration and gravitational gradient derivative ($\frac{d\Gamma}{d\tau}$)". The receiving end utilizes the quadratic term of a Taylor expansion for continuous curvature fitting:

$$\Gamma(\tau + \Delta\tau) \approx \Gamma(\tau) + \frac{d\Gamma}{d\tau} \Delta\tau + \frac{1}{2} \frac{d^2\Gamma}{d\tau^2} (\Delta\tau)^2$$

Without relying on high-frequency pulses, the receiving end can dynamically predict local clock drifts between two discrete pulses, eliminating accumulated phase errors caused by continuous acceleration fields.

Phase perturbations caused by solar winds and interplanetary plasma exhibit common-mode properties for all nodes within the solar system. Through cross-comparison and differential operations among multiple nodes, this background noise can be naturally suppressed without needing individual precise models.

- **Causal Backpressure & Queue Mechanism:** When a sudden extreme change occurs in the flow rate ratio between the source and target ends (e.g., the target end suddenly enters a strong gravity well), causing the target's local causal processing rate to fall behind the upstream input rate, the target end's ingress queue will trigger a critical Backpressure Signal.
 - This signal forces upstream nodes to reduce data transmission throughput.
 - Overloaded event data will be temporarily stored in the "Causal Buffer" of the geometric pipeline, queuing according to the monotonic constraint of $\frac{d\tau}{dt} > 0$.
 - Messages are stretched out during transmission in the latency pipeline, ensuring that no low-flow-rate sub-branch experiences a Buffer Overflow or event loss due to time dilation.

The physical capacity of this buffer is constrained by the discrete-grid information throughput limit J^{cap} ; its precise engineering correspondence is left for subsequent engineering teams to address.

4. Theoretical Completeness and Engineering Boundary Decoupling

- **Theoretical Limit of Precision:** After deducting medium and geometric perturbations, the theoretical resolution of this protocol under a closed geometric loop is constrained by the hardware limits of current optical lattice clocks, capable of reaching $\mathcal{O}(10^{-16} \text{ s})$. This is the hardware resolution upper bound, not the protocol's sole contribution.
- **Engineering Error Budget:** In actual deep space environments, radio signals are interfered by solar wind ionosphere plasma dispersion, solar gravitational field lensing effects, and ephemeris uncertainties (uncompensated uncertainties can reach the $\sim 10^{-11} \text{ s}$ level). In engineering practice, interplanetary handshake pulses employ Multi-frequency Dispersion Compensation and dynamic metric field fitting to periodically correct minor drifts accumulated from sensor numerical integration ($\int \delta\Gamma d\tau$), converging the global alignment error to the nanosecond level ($\mathcal{O}(10^{-9} \text{ s})$).

Chapter 4: Astrophysical Real-Data Calibration and Earth-Mars Interplanetary Derivation Examples

(Note: The $\gamma_{C/A}$ used in this chapter is defined as the **relative** time rate of change between two physical observation endpoints, which must be strictly distinguished from the local time dilation factor Γ_{local} aligned with the topological trunk in Chapter 3.)

1. Regional Calibration Example of the Earth's Surface and GPS Satellites

Considering the combined special and general relativistic effects in low Earth orbit (slower by $7 \mu\text{s/day}$ due to velocity, faster by $45 \mu\text{s/day}$ due to gravitational potential), the measured net drift rate between the Earth's surface reference end and the GPS branch end is $+38 \mu\text{s/day}$. Substituting into the relative rate equation for both endpoints:

$$\gamma_{\text{GPS/Earth}} = \frac{d\tau_{\text{GPS}}}{dt_{\text{Earth}}} \approx 1 + 4.40 \times 10^{-10}$$

This data serves purely as a benchmark proof that "single point rate variations can be measured and restored" within a regional gravity well.

2. Earth-Mars Interplanetary Docking and Worldline Path Integration Calibration

Let Earth's surface (A) and Mars' surface (C) be independent physical branch endpoints. Based on mass and radius parameters, the resting relative rate of change between Earth and Mars is:

$$\gamma_{C/A} = \frac{d\tau_{\text{Mars}}}{dt_{\text{Earth}}} \approx 1 + \frac{GM_E}{R_E c^2} - \frac{GM_M}{R_M c^2} \approx 1 + 5.55 \times 10^{-10}$$

This means a Mars atomic clock is approximately 47.9 μs faster per day than the Earth surface baseline.

This model utilizes the Worldline Continuous Field Integration as its sole dynamic alignment core. All time flow rates are fundamentally continuous, dynamically floating fields. The previous "local static transitive chain laws (such as $\gamma_{C/A} = \gamma_{C/B} \cdot \gamma_{B/A}$)" are merely discrete special cases that naturally degenerate from integration calculations when gravitational potential perturbations approach zero.

In relay communications, satellites only possess "Edge Router" forwarding capabilities and do not act as necessary intermediate nodes for physical causal calculation. The Mars end only needs to receive a single initial handshake pulse from Earth; combined with the dynamic frequency scaling and backpressure mechanisms of Layer Three, it can utilize local triangulation and continuous integration to push the accumulation restoration error close to the hardware resolution limit of optical lattice clocks within a 100-second integration window.

Chapter 5: Microscopic Effective Field Theory Extensions and Falsifiability

This chapter serves as a preliminary framework connecting TCLM with quantum field theory and is not the core argument of this paper.

The microscopic correction terms, dispersion relations, and topological shear operators described herein are preliminary concepts for future research directions; their mathematical completeness and experimental verification are outside the scope of this paper.

[Chapter Scope and EFT Boundary Disclaimer]

The microscopic operators, dispersion relations, and topological shear equations established in this chapter are all positioned as **Effective Field Theory (EFT) phenomenological models** within the TCLM framework. Under subluminal ($v < c$) and macroscopic low-energy limits, this model naturally degenerates into the Lorentz covariance of standard special relativity and the metric of general relativity. Based on the postulate that the Planck length ($\ell_P \approx 1.6 \times 10^{-35} \text{ m}$) and Planck time ($t_P \approx 5.4 \times 10^{-44} \text{ s}$) serve as the absolute minimum units of physical measurement, TCLM acknowledges that spacetime has a discrete grid structure at extreme microscopic levels. The non-linear modification terms in this chapter represent the effective boundary responses of the microscopic discrete grid under macroscopic continuous manifold approximations.

1. Microscopic Dispersion Relation at the Planck Scale

Under ultra-high frequencies and extreme energy states, effective corrections of the microscopic discrete grid are introduced into the dispersion equation:

$$E^2 = p^2 c^2 + m^2 c^4 \pm \xi \left(\frac{E}{E_{\text{Planck}}} \right)^n E^2$$

Here, the coefficient ξ and the energy scale represent the actual pressure exerted by the TCLM "local branch curvature" on microscopic particles, serving as the effective damping of the discrete grid on high-energy wave

packet propagation. If the arrival time differences of high-energy Gamma-Ray Bursts (GRBs) deviate from standard relativistic predictions by a specific threshold, it can directly confirm the non-linear latency effects of this geometric branch under extreme energies.

The order n in the dispersion relation is a free parameter to be constrained by existing high-energy astrophysical observations of Lorentz violation.

2. Corresponding Operator for Quantum Entanglement Phase and Topological Latency

The extreme perturbation latency ϵ_q within the local density matrix is defined as a phase distortion equation:

$$\Delta\Phi_{\text{entangle}} = \frac{2\pi c}{\lambda_c} \left[\int_{\tau_1}^{\tau_2} \left(1 - \frac{1}{\Gamma(\tau)} \right) d\tau + \epsilon_q(\rho_{\text{vacuum}}) \right]$$

Using a distributed local quantum clock network for long-baseline classical pulse differential comparisons, if the decay rate of the entangled EPR correlation function precisely matches the inverse prediction of the trunk dilation factor $\Gamma(\tau)$, it verifies the microscopic phase noise modulation caused by regional vacuum energy density and gravitational gradients.

3. Coupling of Gravitational Fields and Metric Tensors (Topological Information Cutoff and Vacuum Coupling Field Equation)

Formalizing the causal trunk t_{trunk} as the background manifold, the geometric branch manifests as a four-dimensional perturbation tensor:

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}^{\text{branch}}$$

(where $g_{ij}^{\text{branch}} \equiv g_{ij}^{(0)} + h_{ij}^{\text{branch}}$, $\mu, \nu \in \{0, 1, 2, 3\}$, $i, j \in \{1, 2, 3\}$).

Introducing the EFT-based Topological Damping Divergence Operator:

$$\nabla_{\mu} h_{\text{branch}}^{\mu\nu} = \kappa \cdot \text{sgn}(\mathcal{C}) \cdot (J_{\text{info}}^{\nu}) \left[1 - \exp \left(-\frac{\|J_{\text{info}}^{\nu}\|}{J_{\text{cap}}} \right) \right] + Q_{\text{vacuum}}^{\nu}$$

Here, J^{cap} is the information throughput limit that the spacetime discrete grid can absorb through geometric area expansion (event horizon expansion); Q_{vacuum}^{ν} is the energy tensor dissipated into the microscopic background vacuum, ensuring global energy conservation $\nabla_{\mu} T_{\text{total}}^{\mu\nu} = 0$. This equation predicts that when ultra-strong field dynamic collisions exceed the expansion limit, overloaded information flows will impose Information Cutoff Damping and non-linear phase drifts on gravitational waves, providing a precise pathway for empirical falsification via spatial gravitational wave detectors (such as LIGO/LISA).

Chapter 6: Flagship Engineering Application: "Passive Gravitational Wave Detection" Based on the Dual-Pulse Chronometric Network

1. **Spacetime Perturbation Response:** For endpoint networks (such as an Earth-Mars chronometric network) performing long-term dual-pulse dynamic curvature calculations, the derived $\Gamma(t)$ curve not only reflects conventional orbits and gravitational potentials but also produces sensitive transient distortions in response to gravitational waves (Metric Perturbation, $h_{\mu\nu}$) sweeping through local spacetime.

2. **Chronometric Gravitational Wave Sensing:** This mechanism elevates the communication calibration network directly into a distributed "Chronometric Gravitational Wave Detection Network". Through cross-comparison among multiple intergalactic nodes and the differential operation of high-frequency strontium optical lattice clocks, it can effectively filter out common-mode noises like solar wind, offering a new observational pathway that bypasses the arm-length limitations of traditional interferometers.

Chapter 7: Theoretical & Engineering Pressure Tests and Boundary Defenses (Comprehensive Q&A)

ID and Question	Answer and Boundary Defense
Q1: How to explain the permanent time difference in the "Twin Paradox"?	The two atomic clocks undergo different spatial motion trajectories (acceleration and gravitational fields), and the curvature latency accumulated on their branches is inherently unequal. This is a geometric branch curvature difference, not a trunk causal reversal.
Q2: If superluminal speeds are established, is it possible to send signals to the past or achieve causal reversal?	Within the regional geometric observational architecture, absolutely not. Pure geometric acceleration can only cause apparent reverse-order reception of wavefront signals ($\Delta t_{\text{latency}} > 0$); it cannot invert causal vectors within a single manifold. The global topological causal operator is strictly positive-definite ($\text{sgn}(\mathcal{C}) \equiv +1$), and historical coordinates are topologically irreversible, completely sealing off time reversal both geometrically and topologically, making it impossible to send information to the past or trigger a grandfather paradox.
Q3: After matter falls past the event horizon of a black hole, does time stop or reverse?	Absolutely not. The black hole event horizon is merely a geometric gateway where branch curvature reaches its limit; the evolution of matter on the causal trunk is still strictly read forward by observers.
Q4: Does the TCLM causal order destroy tensor covariance?	Not at all. TCLM distinguishes between "geometric observables" and "topological causal order". Local geometric observables (like the clock ticking rate $\Delta t_{\text{latency}}$) completely obey Lorentz covariance and general relativity metric equations (acting as local Effective Field Theories, EFT); whereas t_{trunk} acts solely as the causal sorting scalar of a global topological Directed Acyclic Graph (DAG) and does not interfere with the metric tensor calculations of any regional inertial frame.
Q5: Does the TCLM framework face relativity's limitation of "mass and energy diverging near the speed of light"?	Continuous superluminal (Type A) relies on regional spatial metric manipulation to eliminate geometric resistance of local manifolds; Discrete superluminal (Type B) utilizes topological transitions for displacement, requiring no traversal of continuous space, naturally lifting the energy overflow limit.

ID and Question	Answer and Boundary Defense
Q6: Is TCLM a disguised restoration of Newton's "Absolute Spacetime" or "Ether"?	Absolutely not. Newtonian absolute spacetime presupposes a "universal absolute clock ticking synchronously at all points" and assumes a rigid spatial medium. The t_{trunk} in TCLM is not a clock at all , but purely a unidirectional topological causal operator. TCLM does not provide any global simultaneity slices, nor does it provide any physical resistance to the drifting of mechanical reference frames. All time measured by physical entities (Clock Time) is the geometric path integral over the branch manifold.
Q7: Does the superluminal collapse of quantum entanglement violate the "No-communication Theorem" or cause trunk chaos?	It does not violate it at all. Quantum entanglement within this protocol serves purely as a zero-latency correlation imprint in each pulse unit, functioning in post-facto cross-correlation operations with classical data returns. The dual-pulse relies on two or more independent baselines, not real-time communication via quantum entanglement. Single-sided measurement only yields random eigenvalues and cannot read the remote end's time in real-time; the system utilizes post-return classical data streams to locally perform cross-correlation operations ($R_{AB}(\tau)$) resolving geometric angles, completely independent of superluminal information transfer, forming a strictly decoupled logical closed loop in topology and engineering.
Q8: Will continuous flow rate shifts and traffic surges cause the Dual-Pulse Protocol to collapse?	Not at all. The protocol has introduced mechanisms like Adaptive Pulse Frequency Scaling, Continuous Curvature Quadratic Compensation, and Causal Backpressure Buffering, ensuring data topological integrity and queue safety under continuous acceleration and extreme flow rate differences.

Chapter 8: Conclusion

This paper has successfully deconstructed the mathematical overflow of special relativity under limit speeds into the "apparent reverse-order reception of wavefront signals," establishing the complete architecture of the "topological causal trunk" and "geometric branch curvature" in the TCLM model, while proposing the "Dual-Pulse Protocol" and its underlying variation calculation algorithms as operational standards for engineering implementation.

The model establishes "Spacetime Worldline Continuous Field Integration" as its sole dynamic core. Combining actual measurement data from GPS (+38 $\mu\text{s/day}$) and the Earth-Mars system (47.9 $\mu\text{s/day}$), it validates the feasibility of the protocol to autonomously restore the global causal trunk under dynamic gravity wells and non-inertial frames.

In terms of precision, this paper clearly distinguishes two levels:

- 1. Hardware Resolution Limit:** The local resolution of current optical lattice clocks can reach $\mathcal{O}(10^{-16} \text{ s})$; this is the theoretical upper limit of hardware capability, not a separate contribution of the protocol.
- 2. Engineering Alignment Error Budget:** In real deep-space environments, after multi-frequency dispersion compensation and dynamic metric field fitting, the effective engineering error for

interplanetary global alignment converges to the nanosecond level ($\mathcal{O}(10^{-9} \text{ s})$), which is fully sufficient to meet the rigorous demands of deep-space navigation and communication.

Simultaneously, this protocol can be upgraded into a distributed "Chronometric Gravitational Wave Detection Network," offering a new observation pathway that circumvents the arm-length limitations of traditional interferometers via cross-comparison among multiple intergalactic nodes.

The core contribution of this paper lies not in claiming a complete theory of quantum gravity, but in pointing out: The imaginary numbers and negative time of Lorentz transformations under extreme conditions should be correctly understood as an "Undefined Domain overflow of mathematical models" rather than "backward physical time travel". This epistemological correction provides a brand new topological causal framework for the causal comprehension of superluminal phenomena and deep-space chronometric engineering.

Appendix A: Speculative Extensions under Extreme Speeds

1. Prediction 1: Apparent Time-axis Decoupling & Inversion

- **Observational Phenomenon:** When a superluminal entity (or superluminal wavefront) moves toward the observer, the observer will "first see the arrival image, then receive photons from the starting and middle points."
- **Dual-track Path Mechanism:**
 - **Type A (Continuous Superluminal / Geometric Gliding):** If a physical entity can break the speed of light in continuous space, it will produce a geometric phenomenon isomorphic to a fluid dynamic "supersonic boom". When the object reaches its geometrically closest point, an image appears out of nowhere and splits into two: one moving forward and the other presenting a backward-moving visual afterimage, proving that superluminal speed is merely a restructuring of propagation latency.
 - **Type B (Discrete Superluminal / Topological Transition):** If realized through topological jumps isomorphic to "quantum tunneling," the observer will experience a "Discontinuous Frame": before the old image at point A dissipates, a new image instantly erupts at point B, potentially triggering a single high-energy metric perturbation radiation.
- **Causal Defense:** This phenomenon is an apparent wavefront reversal on geometric branches. On the topological trunk, event settlement remains strictly unidirectional and increasing. **An observer may receive future images early via superluminal means, but can absolutely make no physical intervention on that event.** The grandfather paradox is completely blocked at a topological level.

2. Prediction 2: Read-only Causal Horizon under Extreme Speeds

- When the carrier approaches and exceeds the speed of light, its geometric branch curvature enters an extreme high-pressure zone. Any attempt to transmit "dynamic control commands (Write)" upstream to the main trunk will be rejected by the topological operator.
- Superluminal speed can serve as a carrier for passive observation, but there is absolutely no possibility for it to become a weapon to reverse causality.